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Can the Gravitational Wave Stochastic Background be used as a cosmological probe?

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Abstract

The Stochastic Siren method uses the Stochastic Gravitational Wave Background (SGWB) together with resolved binary black hole mergers to constrain cosmological parameters, has been presented as a new way to measure the rate of expansion of the universe, independent of the current methods. This can prove a valuable way to test the current issues found with the Hubble tension. This project aimed to reproduce the current methods in the literature for using this method and study the source of this reported constraining power. We were able to confirm the increase in constraining power provided by the SGWB, and conclude this is related to the ratio of R_0/H_0 and not each of these individual values. However, our data cannot be completely trusted since our analysis fails to reproduce several well-known characteristics.

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1 Introduction

The discrepancies about the expansion rate of the universe, in particular those relating to the difference between early universe measurements and late universe measurements, have prompted the scientific community to look for alternative measurements that might shed some light on this. Simultaneously, the recent detection of Gravitational waves from the LIGO, VIRGO, KAGRA collaboration (LVK) has provided us with a novel method to attack this problem. Gravitational waves emitted during a merger provide a measure of its distance, so only the redshift of the merger is needed to provide a measure of the expansion rate [1]. While direct observations of the electromagnetic counterpart of the merger could provide a direct measurement of the redshift, cases where this is possible are rare. Instead, most current methods resort to a statistical approach to determine the redshift.

One of the current methods used, the so-called "Spectral Siren", makes use of features of the mass spectrum of the measures to take a statistical approach to determine the redshift, allowing us to measure H_0 using only LVK's data. Recent studies have suggested that adding the data from the Gravitational Waves Stochastic background measured by the LVK collaboration may improve the constraints provided by this method [1, 2].

This project aimed to reproduce this analysis and study the different models involved to try to understand the source of this improvement in constraints. For this, we started by replicating the analysis using purely the "Spectral Siren" Method, for resolved Binary Black Hole (BBH) mergers to which we will refer to as the foreground, separately, we analyzed the results from the gravitational wave background, to which we refer to as background only and lastly we combined both analysis to see the improvement in constraints. All these computations were done using the ICAROGW software [3] that has inbuilt methods for both the Foreground and Background likelihood analysis as well as the population models we wished to test.

2 Theory

2.1 Foreground

When we mention the foreground analysis, we mean the likelihood analysis of the BBH signals resolved and detected. The probability of detection of such an event can be seen as an inhomogeneous Poisson distribution process, where the it's differential merger rate is given by $R_\phi(\phi|\underline{\lambda}, R) = R p(\phi|\underline{\lambda})$, where R is the number of events per unit time throughout the universe, $\underline{\lambda}$ represents the population parameters, ϕ the source parameters and $p(\phi|\underline{\lambda})$ the probability of the an event having source parameters ϕ , knowing the population parameters $\underline{\lambda}$ [1, 2, 4–6]. We now need to consider that only a fraction of the sources are observed and that the source parameters available from LVK are in themselves a probability distribution, returning the following Foreground likelihood distribution [1]:

$$\mathcal{L}(\{d\}|\underline{\lambda}) \propto N^{N_{obs}} e^{-NP_{det}(\underline{\lambda})} \prod_{i=1}^{N_{obs}} \mathcal{L}(\{d_i\}|\underline{\lambda}), \quad (1)$$

where P_{det} is the probability of detection and $N = RT$ is the total number of mergers

happening during a given time interval T .

It is at this stage that ICAROGW [3] provides an easy and intuitive way to calculate this likelihood. Due to its importance to this project, we will include a brief overview of how this is performed. It is possible to study this likelihood not in terms of the individual likelihood of each merger but in terms of the rate of mergers per unit source time and redshift [2, 3, 5, 6]:

$$\mathcal{L}(\{d\}|\underline{\lambda}) \propto e^{N_{exp}(\phi)} \prod_{i=1}^{N_{obs}} \int \mathcal{L}(d_i|\underline{\lambda}, \phi) \frac{dN}{d\phi dt} d\phi dt \propto e^{N_{exp}(\phi)} \prod_{i=1}^{N_{obs}} T \int \mathcal{L}(d_i|\underline{\lambda}, \phi) \frac{dN}{d\phi dt} d\phi \quad (2)$$

This hierarchical likelihood can then be further simplified by assuming a scale-free prior (independent of the rate), meaning that $\pi(N_{exp}) \propto \frac{1}{N_{exp}}$, and returning the following simplified expression:

$$\mathcal{L}(\{d\}|\underline{\lambda}) \propto \prod_{i=1}^{N_{obs}} \frac{\int \mathcal{L}(d_i|\underline{\lambda}, \phi) \frac{dN}{d\phi dt} d\phi}{\int p_{det}(\underline{\lambda}, \phi) \frac{dN}{d\phi dt} d\phi} \quad (3)$$

Now, usually the likelihood of a single event, $\mathcal{L}(d_i|\underline{\lambda}, \phi)$, is not usually accessible; the data available from LVK instead gives us the posterior distribution for each event detected. We are provided with $N_{s,i}$ posterior (PE) samples drawn from: $p(\underline{\lambda}|x_i, \phi) = \mathcal{L}(x_i|\underline{\lambda}, \phi)\pi_{PE}(\underline{\lambda}|\phi)$, where π_{PE} is the prior used to generate the samples in the first place. This can be used to simplify the denominator in the likelihood:

$$\int \mathcal{L}(d_i|\underline{\lambda}, \phi) \frac{dN}{d\phi dt} d\phi \approx \frac{1}{N_{s,i}} \sum_{j=1}^{N_{s,i}} \frac{1}{\pi_{PE}(\underline{\lambda}_{i,j}|\phi)} \frac{dN}{d\phi dt} \bigg|_{i,j} = \frac{1}{N_{s,i}} \sum_{j=1}^{N_{s,i}} w_{i,j} \quad (4)$$

where we define $w_{i,j}$ is the number of mergers happening per unit time. This gives us an important quantity, the effective number of posterior samples, which effectively computes how many values from the posterior samples actually contribute to the evaluation of the number of mergers per unit time:

$$N_{eff,i} = \frac{\left(\sum_{j=1}^{N_{s,i}} w_{i,j}\right)^2}{\sum_{j=1}^{N_{s,i}} w_{i,j}^2} \quad (5)$$

This is the first of the two main quantities used in ICAROGW computations. Now we also need to consider the limitations of the detector, which is done by including the injections file from the search sensitivity estimates of the observing run being studied. These limitations can be determined by the number of expected observations (N_{exp}):

$$N_{exp}(\underline{\lambda}) = T_{obs} \int p_{det}(\underline{\lambda}, \phi) \frac{dN}{d\phi dt} d\phi \quad (6)$$

where p_{det} is given as:

$$p_{det}(\underline{\lambda}, \phi) = \int_{x \text{ is detectable}} \mathcal{L}(x_i|\underline{\lambda}, \phi) dx \quad (7)$$

simplifying N_{exp} as :

$$N_{\text{exp}} \approx \frac{T_{\text{obs}}}{N_{\text{gen}}} \sum_{j=1}^{N_{\text{det}}} \frac{1}{\pi_{\text{inj}}(\theta_j)} \frac{dN}{d\phi dt} \Big|_j = \frac{T_{\text{obs}}}{N_{\text{gen}}} \sum_{j=1}^{N_{\text{det}}} s_j \quad (8)$$

where we are feeding the system with a subset N_{det} of N_{gen} injections as a way to simulate the detection capability of our detector, simplifying the likelihood:

$$\ln[\mathcal{L}(\{x\}|\underline{\lambda})] \approx -\frac{T_{\text{obs}}}{N_{\text{gen}}} \sum_{j=1}^{N_{\text{det}}} s_j + \sum_i^{N_{\text{obs}}} \ln \left(\frac{1}{N_{s,i}} \sum_{j=1}^{N_{s,i}} w_{i,j} \right) \quad (9)$$

This illustrates the basic working of the ICAROGW's Foreground likelihood analysis; here, it is easy to see how the different population models are only contained in $\frac{dN}{d\phi dt}$, making it easier to test the effect of different models.

$$\frac{dN}{d\phi dt} = \frac{dN}{d\phi_s dt_s} \frac{dt_s}{dt_d} \frac{1}{\det(J_{d \rightarrow s})} = \frac{dN}{d\phi_s dt_s} \frac{1}{1+z} \frac{1}{\det(J_{d \rightarrow s})} \quad (10)$$

with $\det(J_{d \rightarrow s}) = \frac{\partial d_L}{\partial z}(1+z)^2$ being the Jacobian for variables between the detector and source frames, leading to our equation becoming :

$$\frac{dN}{d\phi dt dz}(\underline{\lambda}) = \frac{1}{1+z} \frac{1}{\frac{\partial d_L}{\partial z}(1+z)^2} p(z, m_{1,s}, m_{2,s} | \underline{\lambda}) \frac{dV_c}{dz d\Omega} \quad (11)$$

2.2 Background

When considering the Stochastic Gravitational wave background, we can see this as the noise from unresolved sources throughout the universe [1, 2, 7, 8]. This can be described as a spectral energy density amplitude:

$$\Omega_{GW}(f) = \frac{1}{\rho_{\text{crit}}} \frac{d\rho_{GW}}{d\ln(f)} = \frac{f}{\rho_{\text{crit}}} \frac{d\rho_{GW}}{d(f)} \quad , \quad \rho_{\text{crit}} = \frac{3H_0^2 c^2}{8\pi G} \quad (12)$$

Similar to the foreground, all the analysis for this is performed by ICAROGW, so we will again provide a brief overview of how this is done and the important inputs needed. First we want to separate Ω_{GW} into a dimensionless quantity [1]:

$$\Omega_{GW} = \frac{f}{\rho_c c^3} F(\phi, f) \quad (13)$$

where f is the observer's frequency and the $F(\phi, f)$ is the integrated flux of astrophysical sources for a given set of source parameters, ϕ and frequency f :

$$F(\phi, f) = \int p(\phi) \int_0^{z_{\text{max}}} \mathcal{F}(\phi, f) \frac{d\dot{N}^o(\phi, z)}{dz} d\phi dz \quad (14)$$

where $p(\phi)$ is the probability distribution associated with source parameters, $\mathcal{F}(f, z)$ is the source fluence, describing the statistical properties of a single source located at z and $\frac{d\dot{N}^o(\phi, z)}{dz}$ is the number of events per unit observer time per redshift intervals.

The source fluence is the average energy per unit area per unit frequency, we can study this in the source frame by adding a conversion term, $(1+z)^2$, similarly we are

interested in describing this in terms of the energy per solid angle (by adding conversion from energy per unit area to energy per solid angle $(\frac{1}{D_L^2})$). This returns the following fluence equation:

$$\mathcal{F} = \frac{dE}{dAdf} = \frac{(1+z)^2}{D_L^2} \frac{dE_{GW}(\phi, f_s)}{d\Omega df_s} \quad (15)$$

Now the number of events per unit observer time and unit redshift, $\frac{d\dot{N}^o(\phi, z)}{dz}$, can be described in terms of the observer frame comoving event rate density, $R^0(\phi, z)$, and the comoving volume per unit redshift $\frac{dV(z)}{dz}$.

$$\frac{d\dot{N}^o(\phi, z)}{dz} = R^0(\phi, z) \frac{dV(z)}{dz} \quad (16)$$

This can be further simplified by considering the definition of comoving volume :

$$dV_c = \frac{c}{H_0} \frac{(1+z)^2 D_A^2}{E(z)} d\Omega dz \Rightarrow \frac{dV}{dz} = \frac{c}{H_0} \frac{D_c^2}{E(z)} d\Omega \quad (17)$$

Applying the definition of Fluence and the event rate, combined with the knowledge that we want to work on the source frame and not the observer's frame ($R = (1+z)R^0$), allows us now to simplify our expression for the integrated flux:

$$\begin{aligned} F(\phi, f) &= \int p(\phi) \int_0^{z_{max}} \frac{1}{D_c^2} \frac{dE_{GW}(\phi, f_s)}{d\Omega df_s} \frac{R(\phi, z)}{1+z} \frac{c}{H_0} \frac{D_c^2}{E(z)} d\Omega \\ &= \frac{c}{H_0} \int_0^{z_{max}} \frac{R(z)}{(1+z)E(z)} \int p(\phi) \frac{dE_{GW}(\phi, f_s)}{d\Omega df_s} d\phi dz \end{aligned} \quad (18)$$

Now notice that $\int p(\phi) \frac{dE_{GW}(\phi, f_s)}{d\Omega df_s}$ is just the average over all source parameters :

$$F(\phi, f) = \frac{c}{H_0} \int_0^{z_{max}} \frac{R(z)}{(1+z)E(z)} \left\langle \frac{dE_{GW}}{df_s} \right\rangle \bigg|_{f_s} dz \quad (19)$$

Giving us the equation for the Gravitational Wave amplitude:

$$\Omega_{GW}(f) = \frac{f}{\rho_c c^2 H_0} \int_0^{z_{max}} \frac{R(z)}{(1+z)E(z)} \left\langle \frac{dE_{GW}}{df_s} \right\rangle \bigg|_{f_s} dz \quad (20)$$

Once more the dependence of this in Population parameters comes from $E(z)$, that depends on the cosmological parameters and $\left\langle \frac{dE_{GW}}{df_s} \right\rangle = \int p(\phi) \frac{dE_{GW}(\phi, f_s)}{d\Omega df_s}$, where $p(\phi)$ reflects the dependence on the mass and redshift distributions [9].

Experiments looking for the Stochastic Gravitational Wave Background (SGWB), like the ones conducted by the LVK collaboration, aim to measure this amplitude. This can be seen as the dimensionless gravitational wave energy density across the frequency spectrum. SGWB is the result of all the unresolved mergers across the universe, meaning current searches from it assume that we are dealing with a stationary, Gaussian, unpolarized and isotropic background [10]. This can be searched by studying the cross-correlation between different pairs of detectors in the collaboration.

The correlation statistics is given by [2, 10]:

$$\hat{C}_{IJ}(f) = \frac{2}{T_{\text{obs}}} \frac{10\pi^2}{3H_0^2} \frac{f^3}{\gamma_{IJ}(f)} \tilde{S}_I \tilde{S}_J^* \quad (21)$$

where $\tilde{S}_I$ and $\tilde{S}_J^*$ are the Fourier transforms of the detector data and $\gamma_{IJ}(f)$ is the overlap function that describes the geometry of the detectors [2].

The combined cross-correlation estimator describes the information from all the detector pairs:

$$\hat{C}(f) = \frac{\sum_{IJ} \hat{C}_{IJ}(f) \sigma_{IJ}^{-2}}{\sum_{IJ} \sigma_{IJ}^{-2}} \quad (22)$$

where we expect $\langle \hat{C}(f) \rangle = \Omega_{GW}(f)$ and σ_{IJ}^2 is defined as:

$$\sigma_{IJ}^2(f) \approx \frac{1}{2T_{\text{obs}}\Delta f} \left(\frac{10\pi^2}{3H_0^2} \right)^2 \frac{f^6}{\gamma_{IJ}^2(f)} P_I(f) P_J(f) \quad , \quad \sigma^2(f) = \left(\sum_{IJ} \frac{1}{\sigma_{IJ}^2(f)} \right)^{-1} \quad (23)$$

This was calculated assuming a small signal-to-noise ratio and where $P_I(f)$ and $P_J(f)$ are power spectral densities of the detectors I and J, and Δf is the frequency resolution [2, 9, 10].

We can then use this to calculate the likelihood of the SGWB, the background likelihood:

$$\mathcal{L}(\hat{C}|\phi) \propto \exp \left[-\frac{1}{2} \sum_k \left(\frac{\hat{C}(f_k) - \Omega_{GW}(f_k)}{\sigma(f_k)} \right)^2 \right] \quad (24)$$

The combined likelihood can be calculated by just multiplying the two individual likelihood parameters, which implies that we assume that the data used for each analysis is completely independent of the other. This is not exactly true since the resolved mergers we are using do contribute to the background as well; this contribution can be considered negligible since the expected contribution these mergers will have in the background is significantly below current upper limits for the foreground, making this a reasonable approximation.

2.3 Population models

As discussed throughout this section, both the foreground and background likelihood analyses are dependent on the so-called population parameters. These can be separated into 3 types: the rate of events (R_ϕ) and its redshift evolution; the mass distribution of the Black Holes involved in the merger; and the cosmological parameters that influence the evolution of the comoving volume.

The rate of events redshift evolution can be given by the Madau-Dickinson model [9, 11] :

$$\mathcal{R}(z) = R_0 [1 + (1 + z_p)^{-\gamma-\kappa}] \frac{(1 + z)^\gamma}{1 + \left(\frac{1+z}{1+z_p} \right)^{\gamma+\kappa}} \quad (25)$$

The mass distribution probability is given by the Power Law + Peak model [2, 12, 13]:

$$p(m_1|m_{min}, m_{max}, \alpha, \lambda_g, \mu_g, \sigma_g) = [(1-\lambda_g)\mathcal{P}(m_1|m_{min}, m_{max}, -\alpha) + \lambda_g\mathcal{G}(m_1|\mu_g, \sigma_g)] \quad (26)$$

$$p(m_2|m_{min}, m_1, \beta) = \mathcal{P}(m_2|m_{min}, m_1, \beta) \quad (27)$$

where $\mathcal{G}$ is a Gaussian and $\mathcal{P}$ is a Power law, and all the masses refer to the source frame masses.

Lastly, the commoving volume is given by:

$$\frac{dV_c}{dzd\Omega} = \frac{c}{H_0} \frac{D_c^2}{E(z)} \quad (28)$$

These are the models used by [1] and [2], the main papers we aim to reproduce. However, it is in the course of this analysis that we came across different forms of these models that we wished to test.

We have created two modified models to try to better understand the behaviour of some of these constants. The first is a modified mass model whose goal was to have only one parameter with units of mass (an average mass) instead of m_{min} and m_{max} due to our difficulty in finding proper bounds for m_{max} . This is made by defining three new parameters:

$$m_c = \frac{m_{max} + m_{min}}{2}, \quad r = \frac{2m_{min}}{m_{max} + m_{min}}, \quad t = \frac{2\mu_g}{m_{max} + m_{min}} \quad (29)$$

these are now used in the Power Law + Peak model instead of μ_g , m_{min} and m_{max} .

Lastly, it is possible to see that from the models used, R_0 and H_0 should be degenerate. However, this is not entirely clear, especially when studying the SGWB, which has very little constraint power in any of these parameters. As such, we set a new ratio parameter $\frac{R_0}{H_0}$ and use it as our parameter together with H_0 .

3 Data Used

In this project, we used the data from the o3 run of the LVK collaboration, from both the o3a ¹ and o3b ² observing runs. We used 55 resolved mergers, having removed those whose analysis didn't correspond to a BBH system or that had a low signal-to-noise ratio (SNR). We used the injections file corresponding to the search sensitivity estimates expected from this run ³ [12, 13]. The data from the SGWB ⁴ used was taken from the analysis in [10] on the upper limits of the Isotropic Gravitational Wave background.

¹<https://zenodo.org/records/6513631>

²<https://zenodo.org/records/8177023>

³<https://zenodo.org/records/5546676>

⁴<https://dcc.ligo.org/G2001287/public>

4 Results

For this analysis, the simulations were run using the bilby [14] sampler, to analyse the likelihood produced by the ICAROGW [3] package in the Cuillin cluster from the University of Edinburgh Institute of Physics and Astronomy.

4.1 Background Analysis

In the background analysis, the first aspect to check is how good a fit our average results are when compared with the data. This is shown in figure 1 presenting a comparison between the Gravitational Wave Background measured [10], presenting effectively the upper bound that we expect this phenomena to take and the average value of the Ω_{GW} generated by the weighted average of the values obtained by using the posterior distribution of hyperparameter generated by the likelihood analysis.

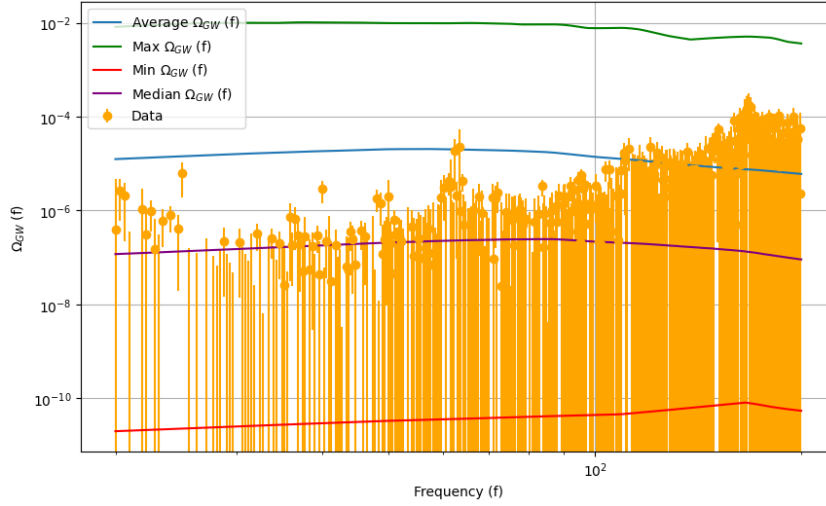


Figure 1: The weighted average Ω_{GW} from the likelihood analysis, its maximum value, minimum value, and median vs the frequency data for the o3 run of the LVK collaboration from [10]. While this data is on a log scale, it is easy to see that while the average data is a good fit for higher frequencies (cropped to 200 Hz), the low-frequency fit is worse. This can be due to the fact that there are more data points at higher frequencies, causing it to shift towards those points. However, since the median data seems to fit this data well, we will not worry about this.

The full hypermeter fittings can be found in Appendix B. Focusing only on H_0 , R_0 and Ω_m we can see that the background has close to no constraining power for these values, as it can be seen in figure 2a. The same is true for the rest of the parameters, as can be seen in figure 7.

4.2 Combined Analysis

When studying the effects of the combined background and foreground likelihood, it's important to compare this with the foreground analysis alone, since these are the results we are aiming to improve with this new analysis. This analysis has not been successful

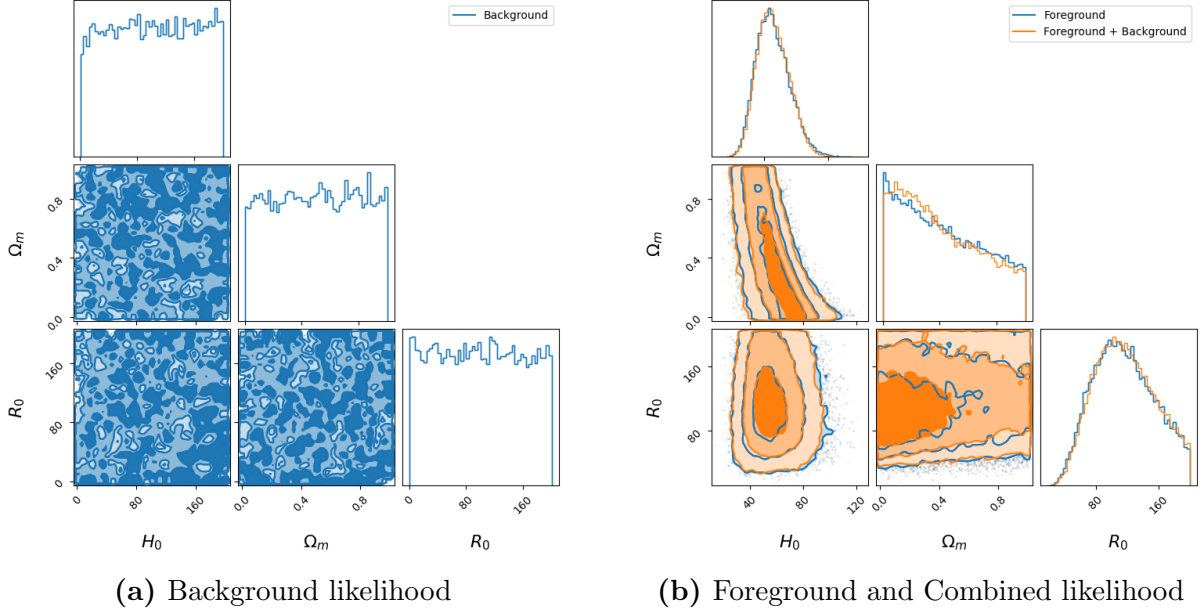


Figure 2: Likelihood analysis for H_0 , R_0 and Ω_m .

since even the foreground analysis has failed to reproduce the expected features for this data, as can be seen when comparing figure 11 in Appendix C.1 that presents the expected features from the literature with figure 6, in particular the most likely value for R_0 is an order of magnitude bigger than the expected one and m_{max} is not well bounded. Despite this, it is still possible to see in Figure 2b that the combined likelihood produces a slightly better value for H_0 , as expected from [1]. Similarly, the distributions for H_0 and Ω_m are very similar to the ones found in [2], as can be seen when comparing figure 2b and figure 13. However, the results found for H_0 provide much smaller improvements than the ones expected from current literature, as shown by comparing figure 3 and the results from [1] shown in figure 12.

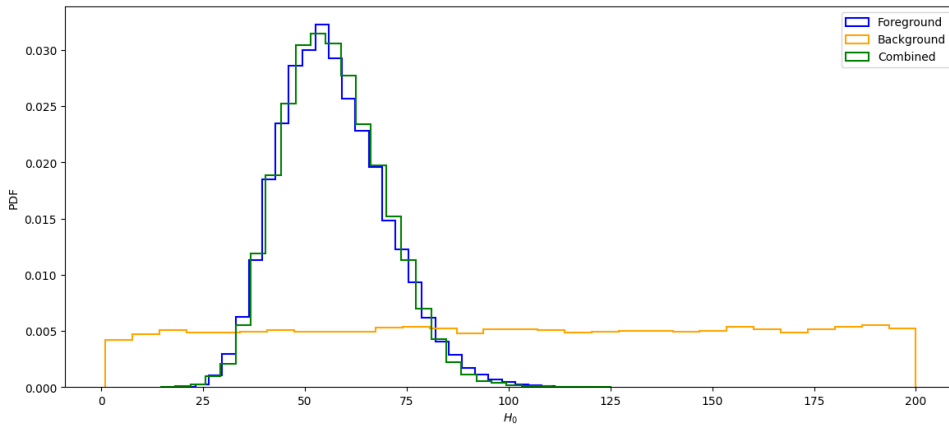


Figure 3: Plot of the Posterior distribution for H_0 from the Foreground, Background and Combined distribution analysis.

4.3 Modified Models

To study the discrepancies in the mass values we have been getting, as well as studying the true constraining power of the background, we also performed likelihood analysis using the modified models described in Section 2.3.

Firstly, to try to understand our problems with the mass values, we used the modified mass model described in equation 29. Figure 4 shows how this affects both the background and foreground likelihoods. However, this seems to indicate very unphysical values for both values of H_0 and Ω_m , indicating something is wrong with our models.

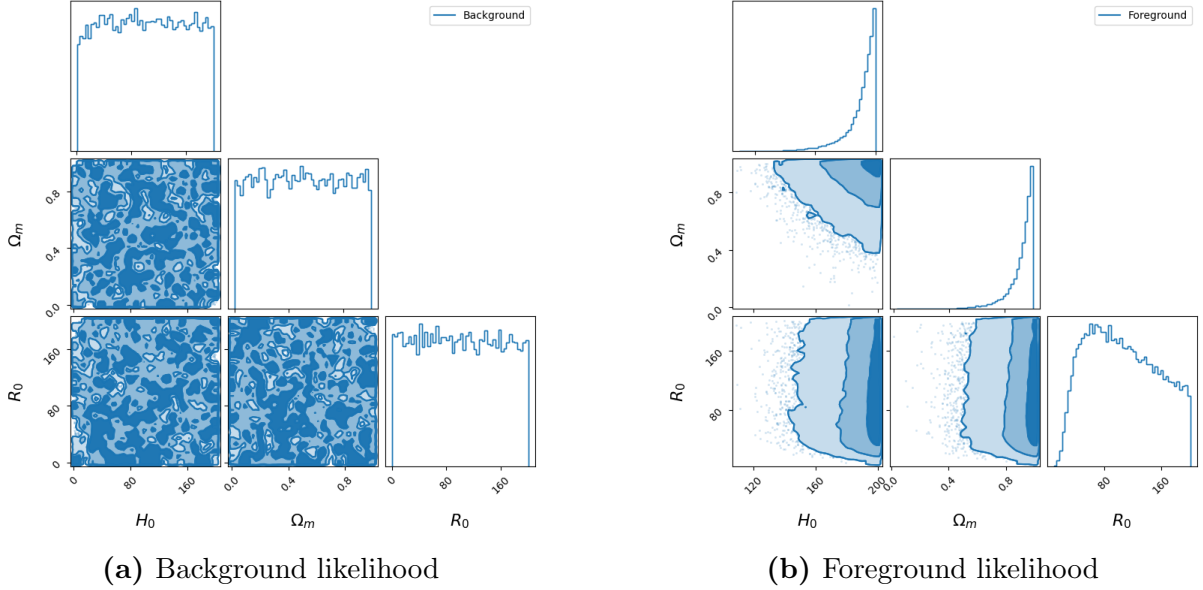


Figure 4: Modified mass likelihood analysis

To test the background likelihood constraining power, we also used a modified ratio model to test if this combination showed a better bound. Figure 5 shows how the value of $\frac{R_0}{H_0}$ is relatively well constrained while the value of H_0 is not constrained at all. This seems to indicate that the slight preference of the background for high values of H_0 and small values of R_0 is in fact a preference for small values of this ratio.

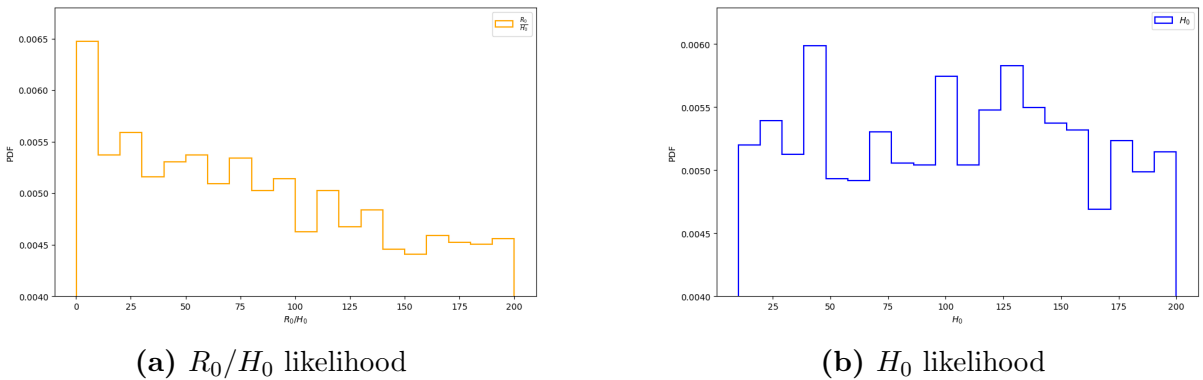


Figure 5: Background Likelihood values for the ratio of $\frac{R_0}{H_0}$

5 Conclusion and Next Steps

This analysis aimed to study how the SGWB, combined with resolved BBH mergers, provides a constraint on H_0 , the rate of expansion of the universe. We were able to find a slight improvement in constraints by using this method, in accordance with the current literature. Furthermore, we were able to confirm that the background presents a considerably better constraint for the value of the ratio R_0/H_0 than R_0 or H_0 separately. However, our results cannot be entirely trusted since they don't reproduce several well-known characteristics of the foreground analysis; furthermore, the results from the modified mass seem to indicate something fundamentally wrong with the mass distribution.

Further work should focus on rooting out the source of these discrepancies in the data and completing this analysis in a way that can be completely trusted. The main source for these discrepancies seems to be related to the priors used in our analysis. While the .h5 files containing the data for the different gravitational wave events do contain a prior distribution, this is only a sample distribution and doesn't correspond to the full number of events in the distribution. To deal with this, the priors used are generated by us following the description of the priors provided in [12]. However, the poor results in our analysis seem to indicate this is not appropriate and that the results are very sensitive to the priors, as such, the next steps in this analysis will involve fixing this and making sure we are using appropriate priors. Adding to this, with the recent release of the data for the first half of the o4 run, it would be worthwhile to perform the same analysis on this new data set and learn how this new data affects the current results.

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A Population Parameters and Priors

This section presents the different population parameters associated to the models described in section 2.3 as well as the priors used in the analysis that produced the results presented here.

Parameter	Prior
Power Law + Peak	
λ_g	(0,1)
μ_g	(20,50)
σ_g	(0.4,10)
α	(1.5,4)
β	(-4,12)
m_{min}	(2,10)
m_{max}	(50,200)
Modified mass	
m_c	(1,250)
r	(0,1)
t	(0,1)
Madau-Dickinson	
γ	(0,12)
κ	(0,6)
z_p	(0,4)
R_0	(0,200)
Cosmological Parameters	
H_0	(10,200)
Ω_m	(0,1)
Rate Model	
$\frac{R_0}{H_0}$	(0, 200)

Table 1: Model Parameters

B Corner Plots

Corner plots for all the models used in this analysis.

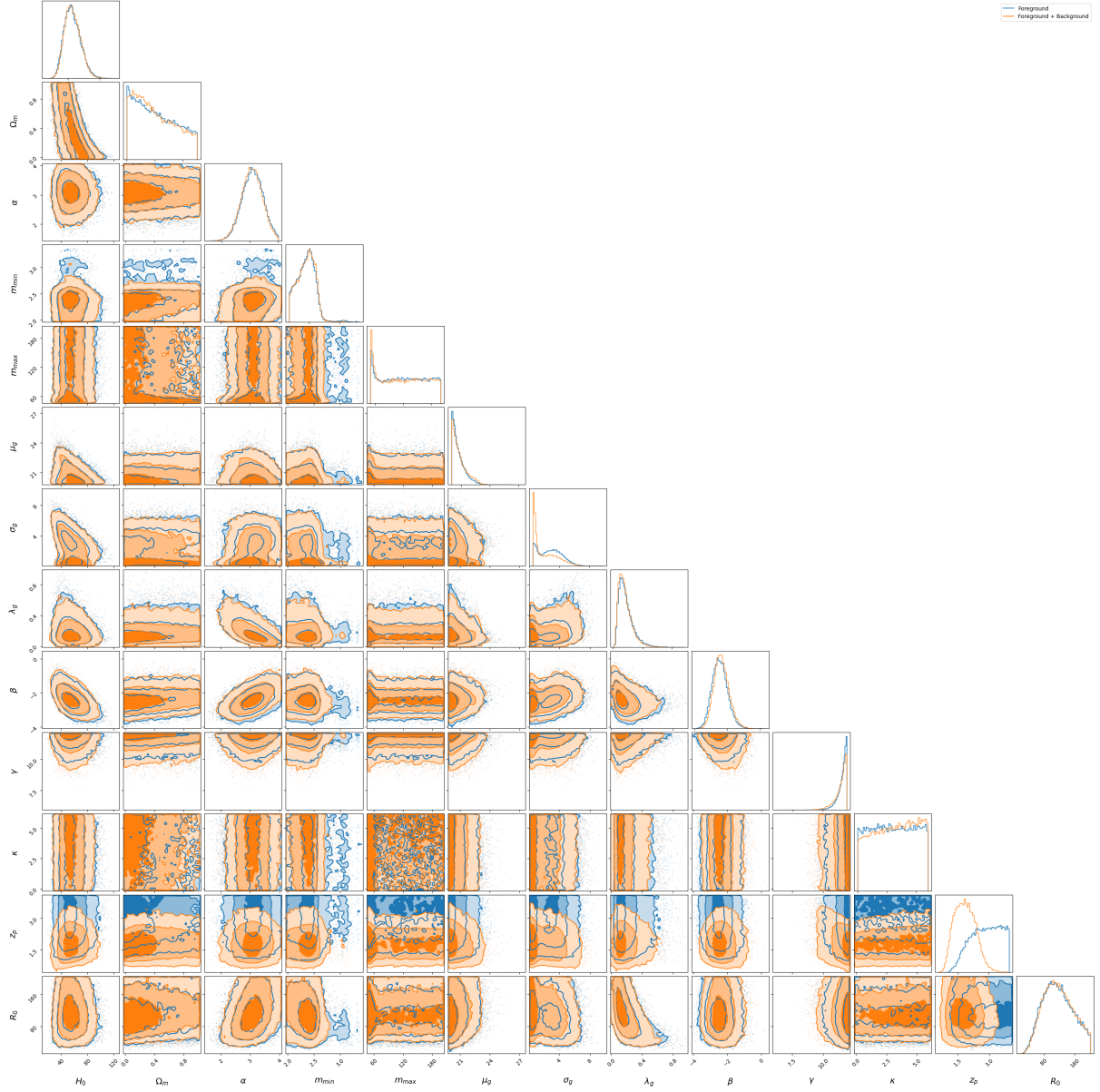


Figure 6: Foreground and combined (Foreground+Background) likelihood corner plot for the Power Law + Peak and Madau-Dickinson models.

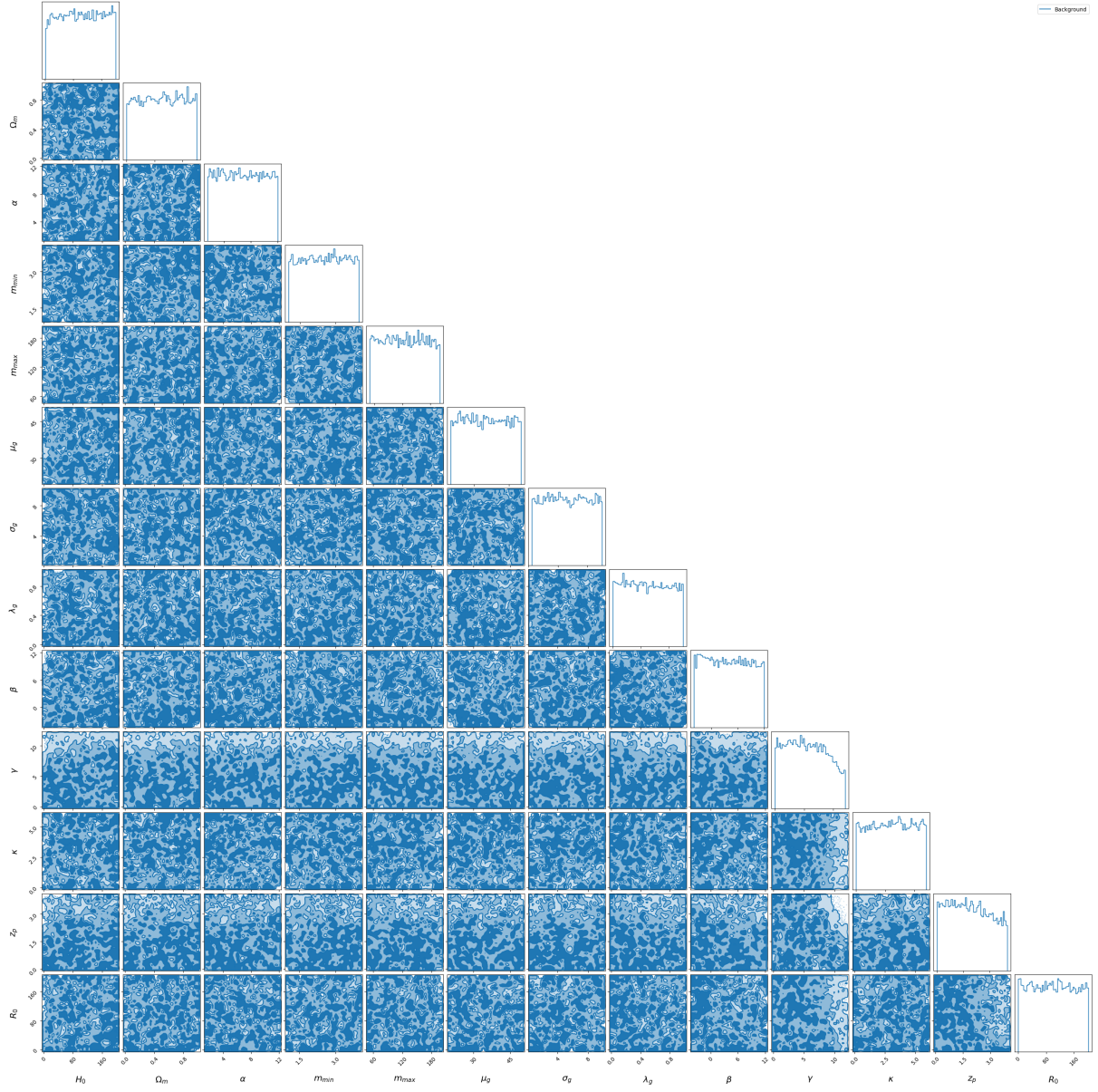


Figure 7: Background likelihood corner plot for the Power Law + Peak and Madau-Dickinson models.

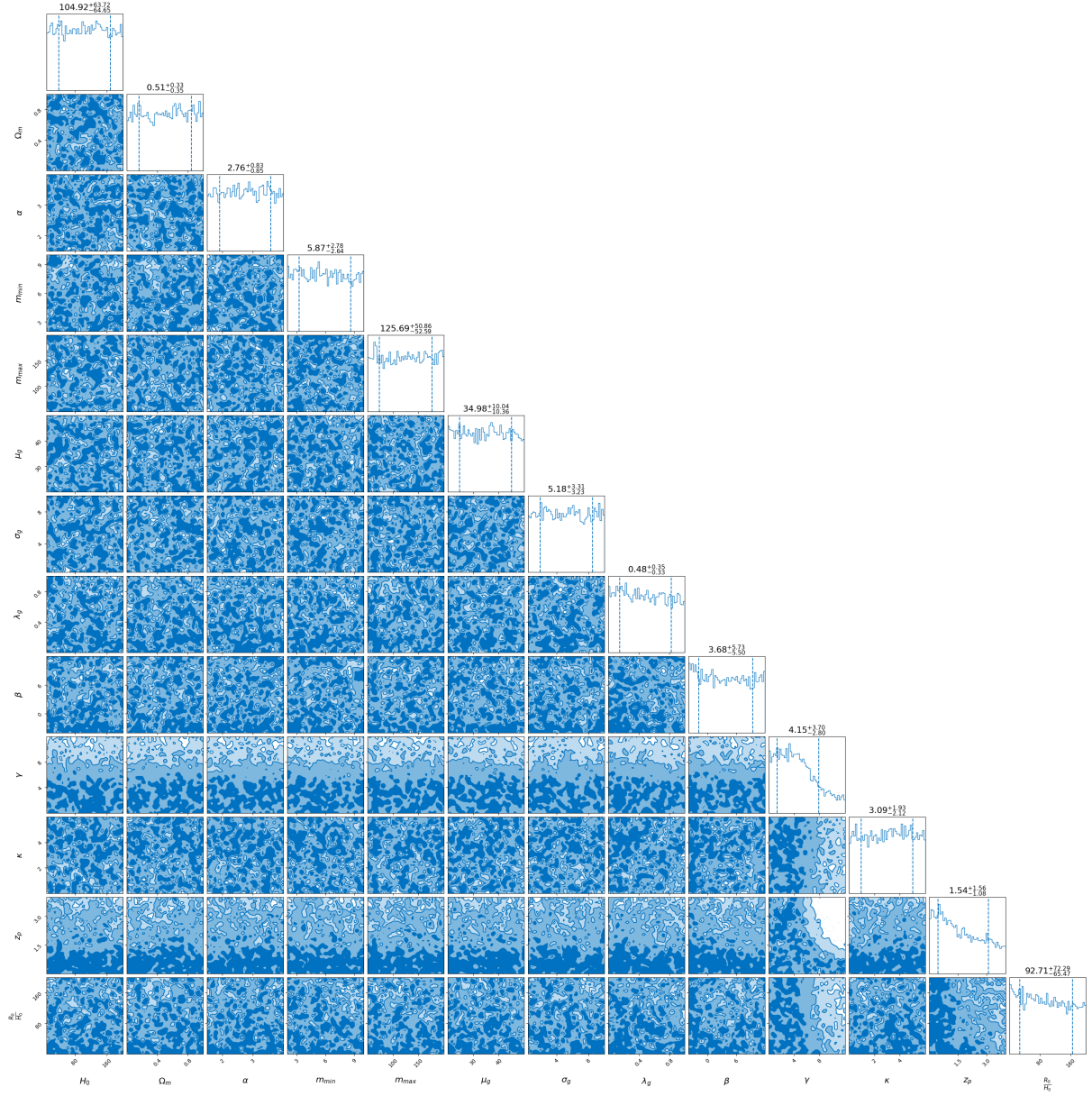


Figure 8: Background likelihood corner plot for the modified ratio model.

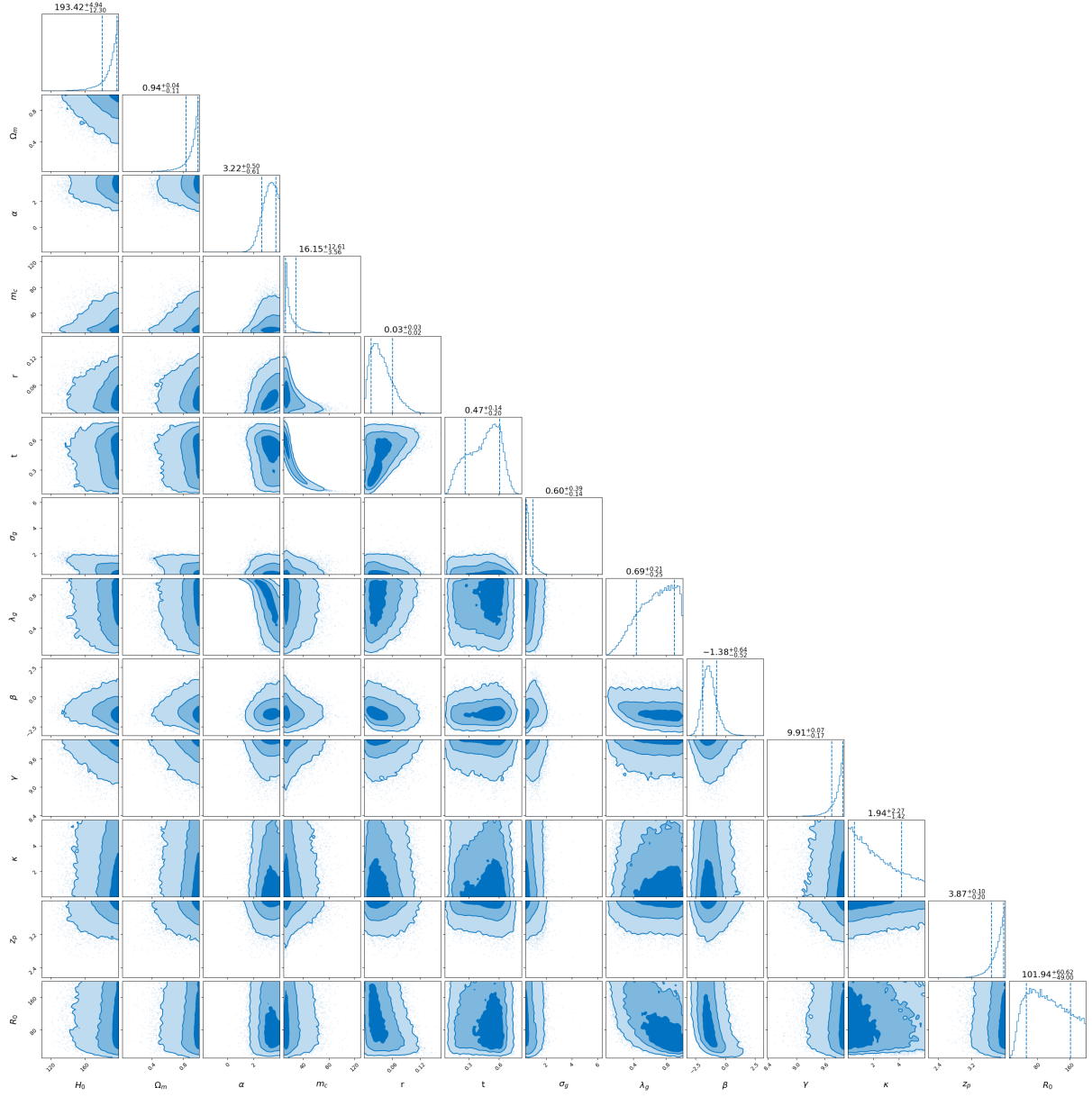


Figure 9: Foreground likelihood corner plot for the modified mass model.

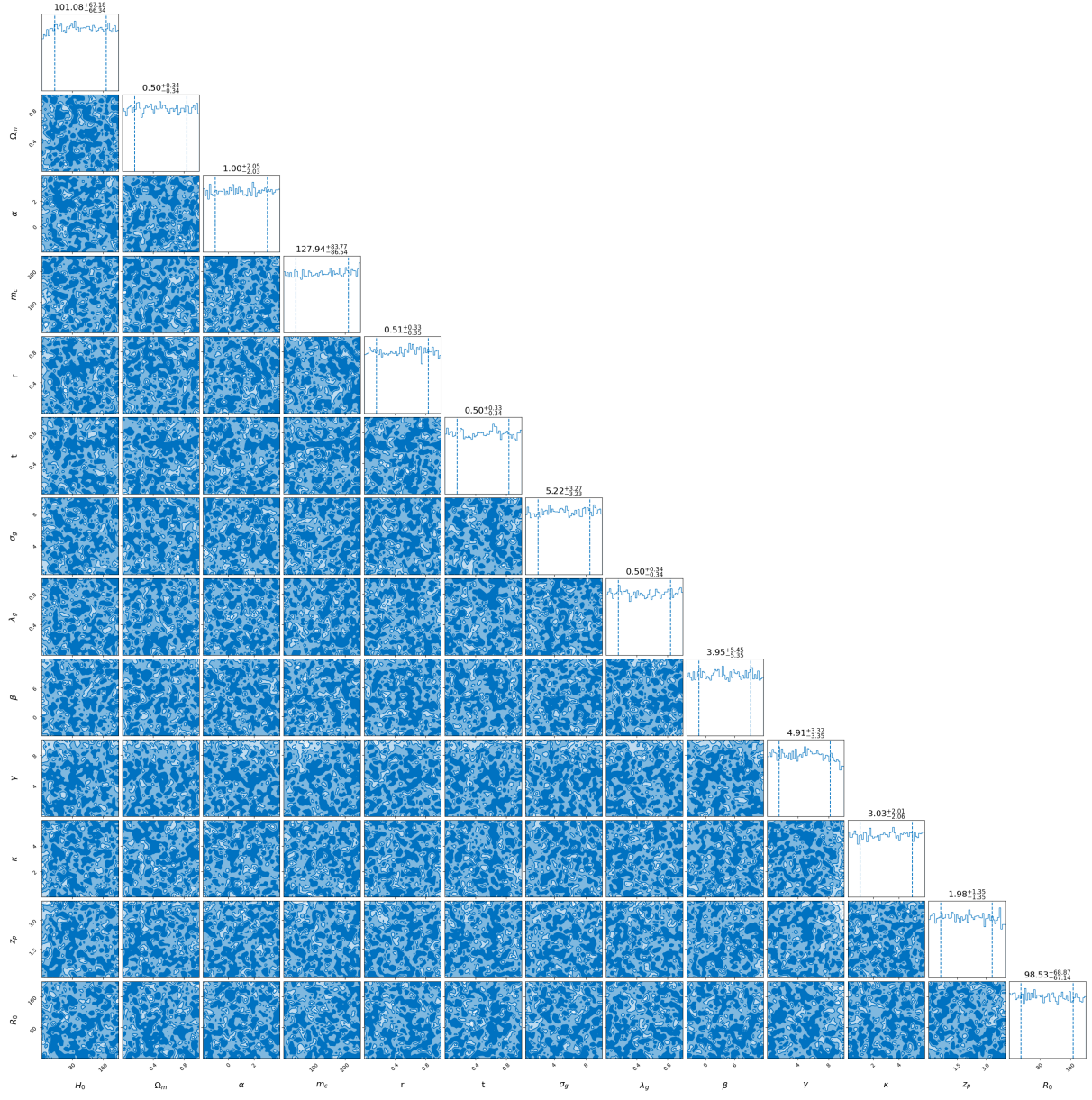


Figure 10: Background likelihood corner plot for the modified mass model.

C Comparison Data

Some plots from different articles, this analysis should be able to reproduce. The goal of these is to provide a useful comparison when analysing how trustworthy our analysis is.

C.1 o3 Foreground Analysis

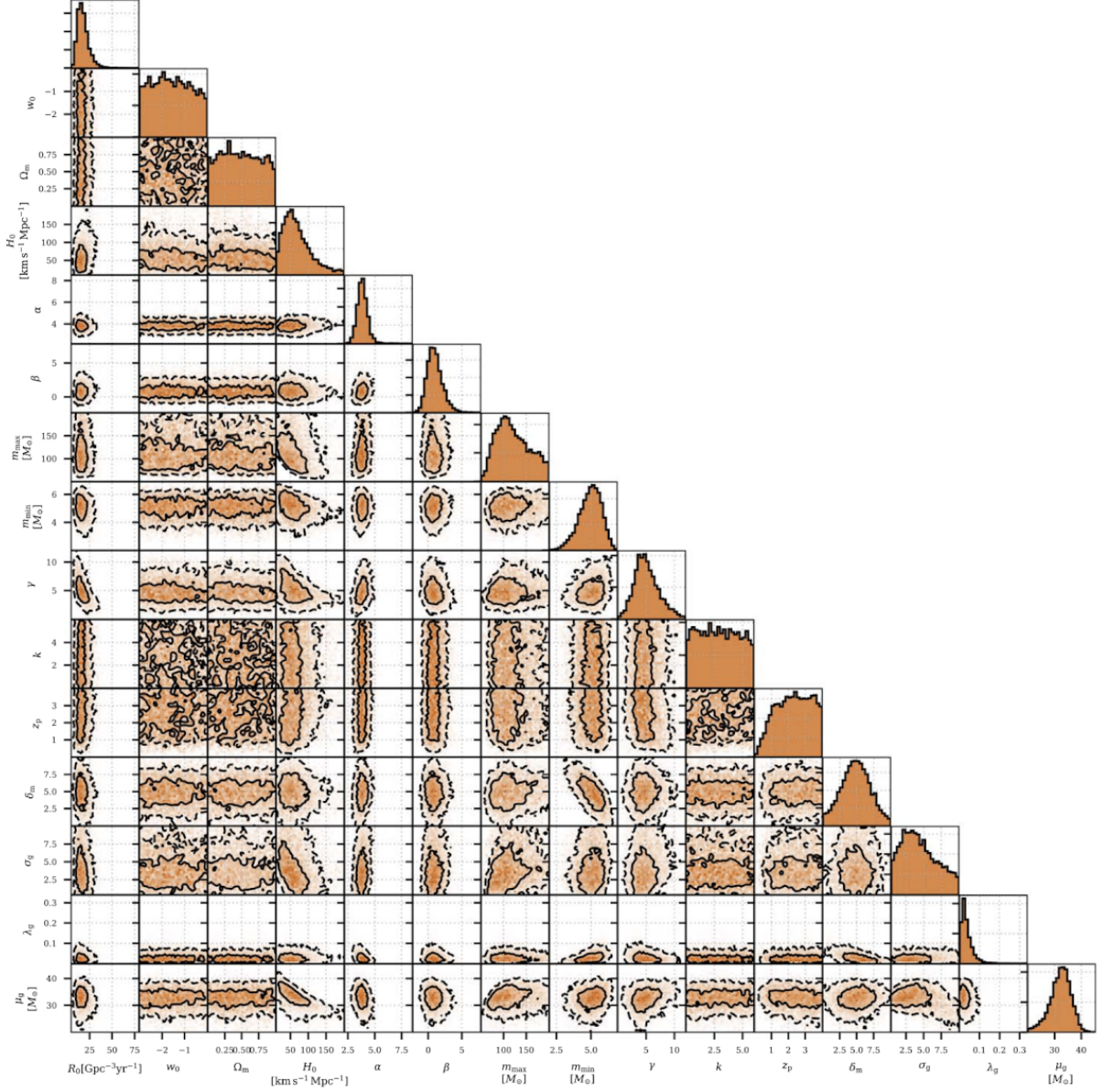


Figure 11: Plot of the Power Law + Peak and cosmological parameters for BBHs from [12]

C.2 Stochastic Siren effect on H_0

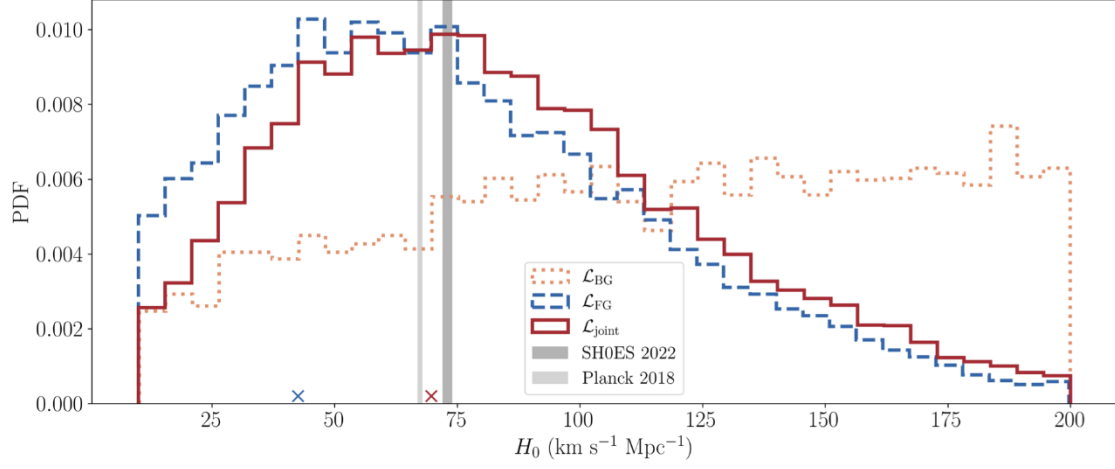


Figure 12: Plot of the Posterior distribution from the Foreground, Background and Joint distribution analysis performed in [1] for H_0 . It shows a clear preference for higher values of H_0 , can this be confirmed by our analysis?

C.3 Corner Plot from Foreground and Combined analysis

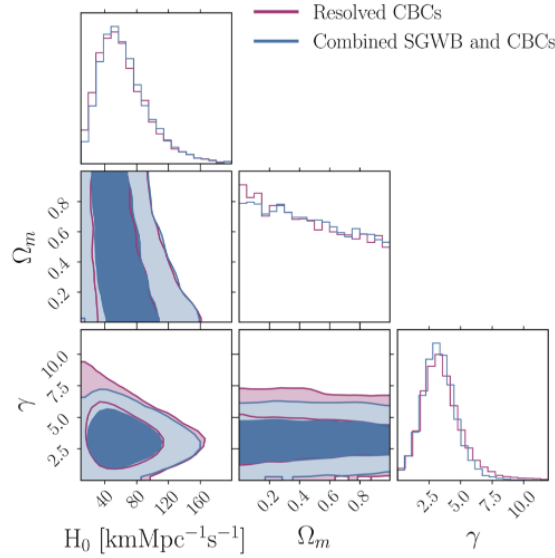


Figure 13: Plot of the likelihood for 59 resolved BBH and for the combined analysis with the background for H_0 , the Madau evolution rate γ and the matter density parameter Ω_m as performed in [2] .